Symmetric and Skew-Symmetric Tensors, Invariants and Decompositions

Part 5 of the OT Science Formula Completion Workbook

OT Science Notes

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Contents

1 Symm and Skew-Symm Tensors, Invariants n' Decomps

Second-order tensors appear naturally as stress tensors, strain tensors, inertia tensors, conductivity tensors, deformation gradients and vector gradients. Their symmetric and skew-symmetric parts often carry different physical meanings.

1.1 Symmetric and Skew-Symmetric Tensors

A tensor is symmetric if transposing it does not change it. A tensor is skew-symmetric if transposing it changes its sign. In three dimensions, a skew-symmetric tensor has only three independent components and can be represented by an axial vector.

Symmetry Conditions

$$S_{ij} = S_{ji} \qquad \text{symmetry.}$$

$$W_{ij} = -W_{ji} \qquad \text{skew-symmetry.}$$

$$W_{ii} = 0.$$

$$A_{ij} = S_{ij} + W_{ij}.$$

$$S_{ij} = \frac{1}{2}(A_{ij} + A_{ji}), \qquad W_{ij} = \frac{1}{2}(A_{ij} - A_{ji}).$$

1.2 Connection with Vector Gradient

If $\mathbf{v}$ is a velocity field, then $\nabla\mathbf{v}$ can be decomposed into symmetric and skew-symmetric parts. The symmetric part is related to strain rate, while the skew-symmetric part is related to local rotation.

Gradient Decomposition

$$L_{ij} = \partial_j v_i.$$

$$D_{ij} = \frac{1}{2}(L_{ij} + L_{ji}) \qquad \text{symmetric rate-of-deformation tensor.}$$

$$\Omega_{ij} = \frac{1}{2}(L_{ij} - L_{ji}) \qquad \text{skew spin tensor.}$$

$$L_{ij} = D_{ij} + \Omega_{ij}.$$

$$\omega_i = (\nabla \times \mathbf{v})_i = \varepsilon_{ijk}\partial_j v_k.$$

1.3 Principal Invariants of a Second-Order Tensor

Invariants do not change under a rotation of coordinates. They describe intrinsic properties of the tensor.

Principal Invariants

For a 3×3 tensor $\mathbf{A}$,

$$I_1 = \mathrm{tr}(A) = A_{ii}.$$

$$I_2 = \frac{1}{2}\left[(\mathrm{tr}\, A)^2 - \mathrm{tr}\big(A^2\big)\right].$$

$$I_3 = \det(A).$$

The characteristic polynomial may be written as

$$\det(\lambda I - A) = \lambda^3 - I_1\lambda^2 + I_2\lambda - I_3.$$

Equivalently,

$$\det(A - \lambda I) = -\lambda^3 + I_1\lambda^2 - I_2\lambda + I_3.$$

1.4 Deviatoric and Spherical Parts

Many physical tensors can be decomposed into a mean or spherical part and a deviatoric part. In stress analysis, the spherical part is associated with hydrostatic pressure, while the deviatoric part is associated with shape distortion.

Deviatoric Decomposition

$$\mathrm{sph}\,(A)_{ij} = \frac{1}{3}A_{kk}\delta_{ij}.$$

$$A'_{ij} = \mathrm{dev}((A))_{ij} = A_{ij} - \frac{1}{3}A_{kk}\delta_{ij}.$$

$$A_{ij} = \frac{1}{3}A_{kk}\delta_{ij} + A'_{ij}.$$

$$A'_{kk} = 0.$$

For a stress tensor σ_{ij},

$$p = -\frac{1}{3}\sigma_{kk}, \qquad s_{ij} = \sigma_{ij} - \frac{1}{3}\sigma_{kk}\delta_{ij}.$$

1.5 Invariants of Deviatoric Tensors

For a deviatoric tensor A', the first invariant is zero. The remaining invariants are important in mechanics.

Deviatoric Invariants

$$J_1 = \mathrm{tr}\big(A'\big) = 0.$$

$$J_2 = \frac{1}{2}A'_{ij}A'_{ij}.$$

$$J_3 = \det\big(A'\big).$$

For stress deviator s_{ij}, one common convention is

$$J_2 = \frac{1}{2} s_{ij} s_{ij}.$$

Worked Example: Tensor Decomposition

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & 0 & -1 \\ 5 & 2 & 1 \end{pmatrix}.$$

Then

$$S = \frac{1}{2}(A + A^T), \qquad W = \frac{1}{2}(A - A^T).$$

Compute

$$S = \begin{pmatrix} 2 & 5/2 & 4 \\ 5/2 & 0 & 1/2 \\ 4 & 1/2 & 1 \end{pmatrix}, \qquad W = \begin{pmatrix} 0 & -3/2 & -1 \\ 3/2 & 0 & -3/2 \\ 1 & 3/2 & 0 \end{pmatrix}.$$

Check that $S^T = S$, $W^T = -W$, and $A = S + W$.

Quick Drills and Practice: Tensor Decompositions and Invariants

Level A: Symmetry checks

1. Is $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ symmetric?
2. Is $\begin{pmatrix} 0 & -4 \\ 4 & 0 \end{pmatrix}$ skew-symmetric?
3. Complete: $S_{ij} - S_{ji} =$ __________.
4. Complete: $W_{ij} + W_{ji} =$ __________.
5. Complete: $W_{ii} =$ __________.
6. Complete: $A_{ij} =$ __________.
7. Complete: $S_{ij} =$ __________.
8. Complete: $W_{ij} =$ __________.

Level B: Decomposition

1. Decompose $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$ into symmetric and skew-symmetric parts.
2. Decompose $A = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{pmatrix}$. What do you notice?
3. For $\mathbf{v} = (-y, x, 0)$, compute $\nabla\mathbf{v}$, its symmetric part and its skew part.
4. For $\mathbf{v} = (ax, by, cz)$, compute $\nabla\mathbf{v}$, its trace and its symmetric/skew parts.

Level C: Invariants

1. Compute I_1, I_2, I_3 for $A = \text{diag}(1, 2, 3)$.
2. Compute I_1, I_2, I_3 for $A = \text{diag}(a, b, c)$.

3. Compute I_1, I_2, I_3 for $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.
4. Write the characteristic polynomial of a 3×3 tensor using I_1, I_2, I_3.
5. If a tensor has eigenvalues $\lambda_1, \lambda_2, \lambda_3$, express I_1, I_2, I_3 in terms of the eigenvalues.

Level D: Deviatoric tensors

1. Find the deviatoric part of $A = \mathrm{diag}(4, 1, 1)$.
2. Find the deviatoric part of $A = \mathrm{diag}(6, 3, 0)$.
3. Prove that $A'_{kk} = 0$.
4. For $A = \mathrm{diag}(4, 1, 1)$, compute $J_2 = \frac{1}{2} A'_{ij} A'_{ij}$.
5. For $\sigma = \mathrm{diag}(100, 50, 25)$, compute the hydrostatic part and deviatoric part.

Level E: Challenge

1. Show that the trace of a skew-symmetric tensor is zero.
2. Show that $x_i W_{ij} x_j = 0$ for any skew-symmetric tensor W_{ij}.
3. Show that $x_i S_{ij} x_j$ depends only on the symmetric part of a tensor.
4. If A is symmetric, explain why its eigenvalues are real in ordinary Euclidean space.
5. Explain why invariants are useful when comparing the same tensor in two different coordinate frames.